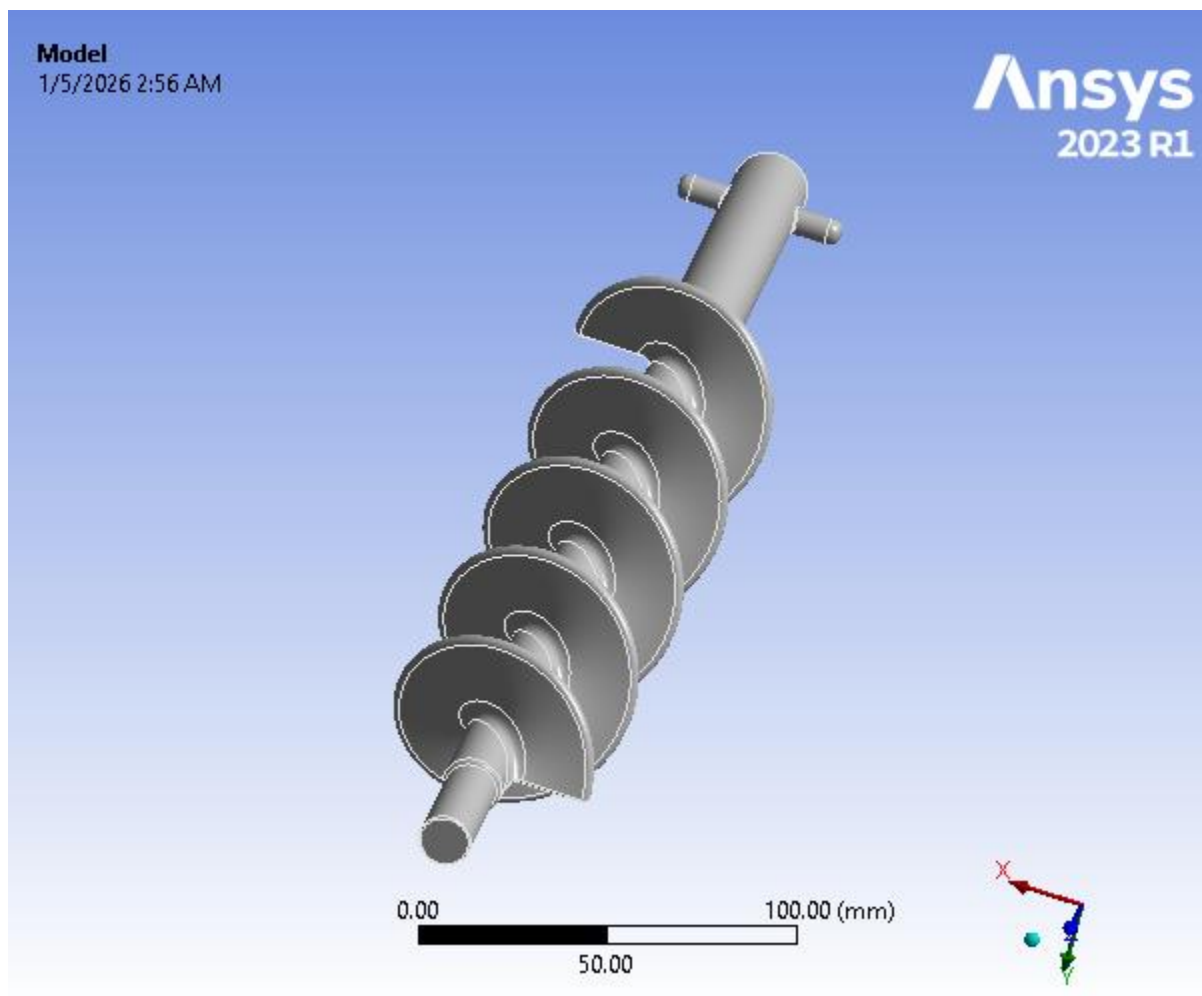




Project*

First Saved	Monday, January 5, 2026
Last Saved	Monday, January 5, 2026
Product Version	2023 R1
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2

Model (A4) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	<i>Geometry Import (A3)</i>
State	Solved
Definition	
Source	C:\Users\user1\Desktop\VFFS\Auger.SLDprt
Type	SOLIDWORKS
Basic Geometry Options	

Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\user1\Desktop\VFFS\Auger.SLDPRT
Type	SOLIDWORKS
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	62.614 mm
Length Y	62.604 mm
Length Z	441. mm
Properties	
Volume	1.4702e+005 mm ³
Mass	1.1762 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	2917486
Elements	2069702

Mesh Metric	Skewness
Min	1.2068e-003
Max	0.58098
Average	0.12096
Standard Deviation	0.10332
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	<i>Auger</i>
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Stainless Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	

Length X	62.614 mm
Length Y	62.604 mm
Length Z	441. mm
Properties	
Volume	1.4702e+005 mm ³
Mass	1.1762 kg
Centroid X	-1.9829e-004 mm
Centroid Y	-0.46815 mm
Centroid Z	95.371 mm
Moment of Inertia Ip1	16722 kg·mm ²
Moment of Inertia Ip2	16724 kg·mm ²
Moment of Inertia Ip3	204.08 kg·mm ²
Statistics	
Nodes	2917486
Elements	2069702
Mesh Metric	Skewness
Min	1.2068e-003
Max	0.58098
Average	0.12096
Standard Deviation	0.10332

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Mesh

TABLE 8
Model (A4) > Mesh

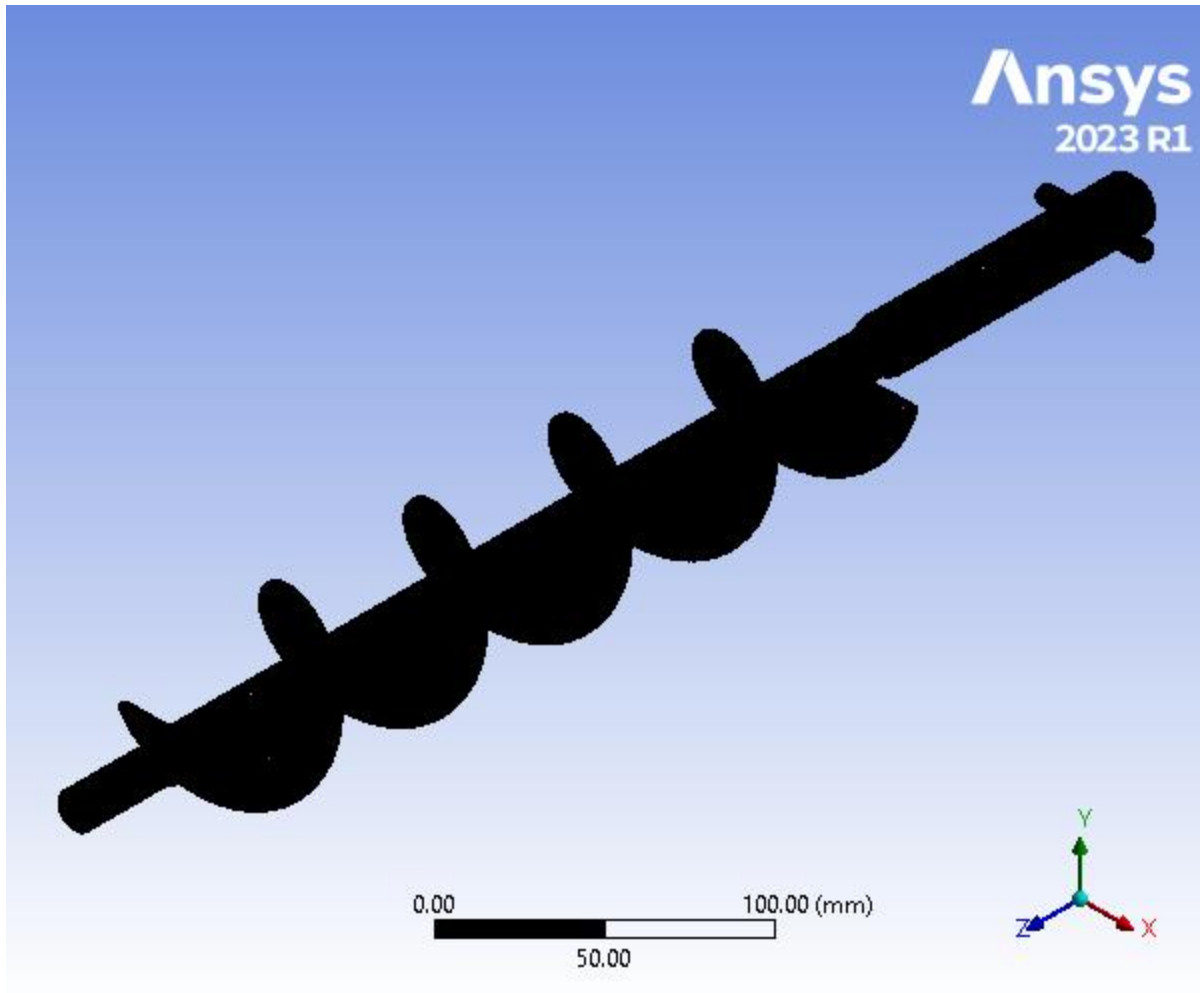
Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	1.0 mm
Sizing	
Use Adaptive Sizing	Yes
Resolution	7
Mesh Defeaturing	Yes
Defeature Size	5. mm
Transition	Slow
Span Angle Center	Fine
Initial Size Seed	Assembly
Bounding Box Diagonal	449.8 mm
Average Surface Area	1313.7 mm ²
Minimum Edge Length	6.2658e-002 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	High
Mesh Metric	Skewness
Min	1.2068e-003
Max	0.58098
Average	0.12096
Standard Deviation	0.10332
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	2917486

Elements	2069702
Show Detailed Statistics	No

TABLE 9
Model (A4) > Mesh > Mesh Controls

Object Name	<i>Patch Independent</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Suppressed	No
Method	Tetrahedrons
Algorithm	Patch Independent
Element Order	Use Global Setting
Advanced	
Defined By	Max Element Size
Max Element Size	Default (1.0 mm)
Feature Angle	30.0°
Mesh Based Defeaturing	Off
Refinement	Proximity and Curvature
Min Size Limit	1.0 mm
Num Cells Across Gap	Default
Curvature Normal Angle	Default
Smooth Transition	Off
Growth Rate	Default
Minimum Edge Length	6.2658e-002 mm
Write ICEM CFD Files	No

FIGURE 1
Model (A4) > Mesh > mesh



Static Structural (A5)

TABLE 10
Model (A4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 11
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	

Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Off
Output Controls	
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\user1\Desktop\VFFS\auger ansys_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes

Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	nmm

TABLE 12
Model (A4) > Static Structural (A5) > Loads

Object Name	Moment	Fixed Support
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	3 Faces	2 Faces
Definition		
Type	Moment	Fixed Support
Define By	Components	
Coordinate System	Global Coordinate System	
X Component	0. N·mm (ramped)	
Y Component	0. N·mm (ramped)	
Z Component	19000 N·mm (ramped)	
Suppressed	No	
Behavior	Deformable	
Advanced		
Pinball Region	All	

FIGURE 2
Model (A4) > Static Structural (A5) > Moment

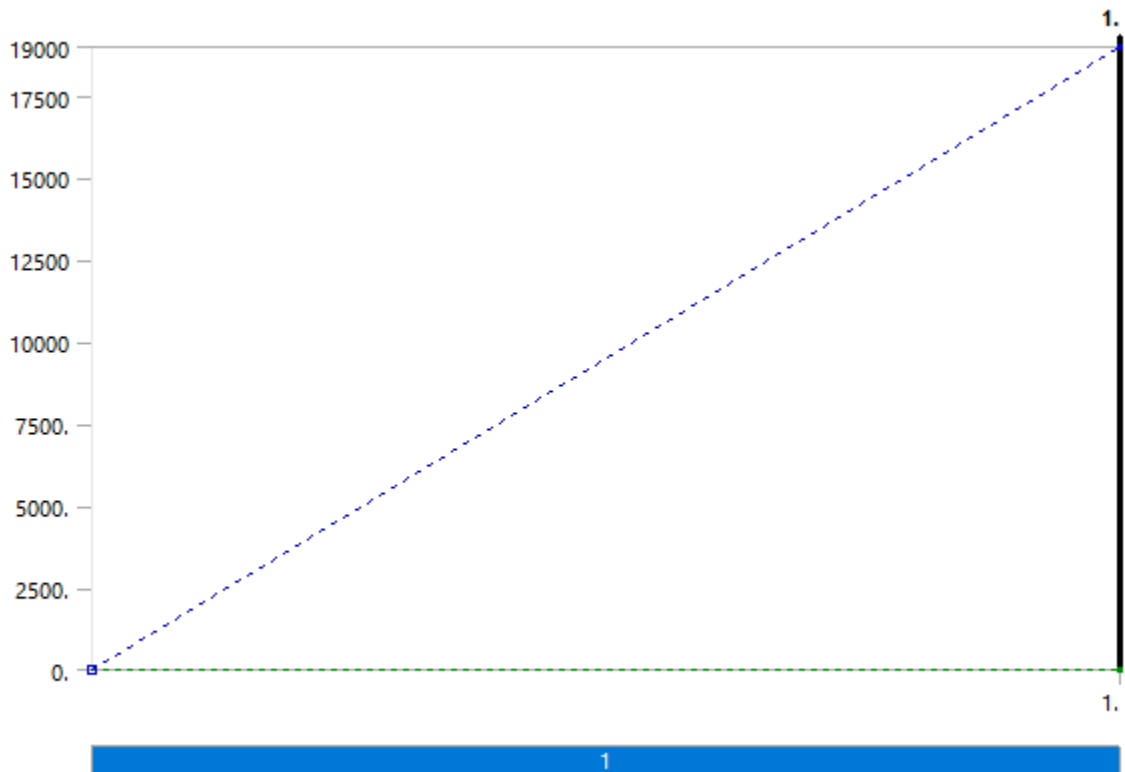
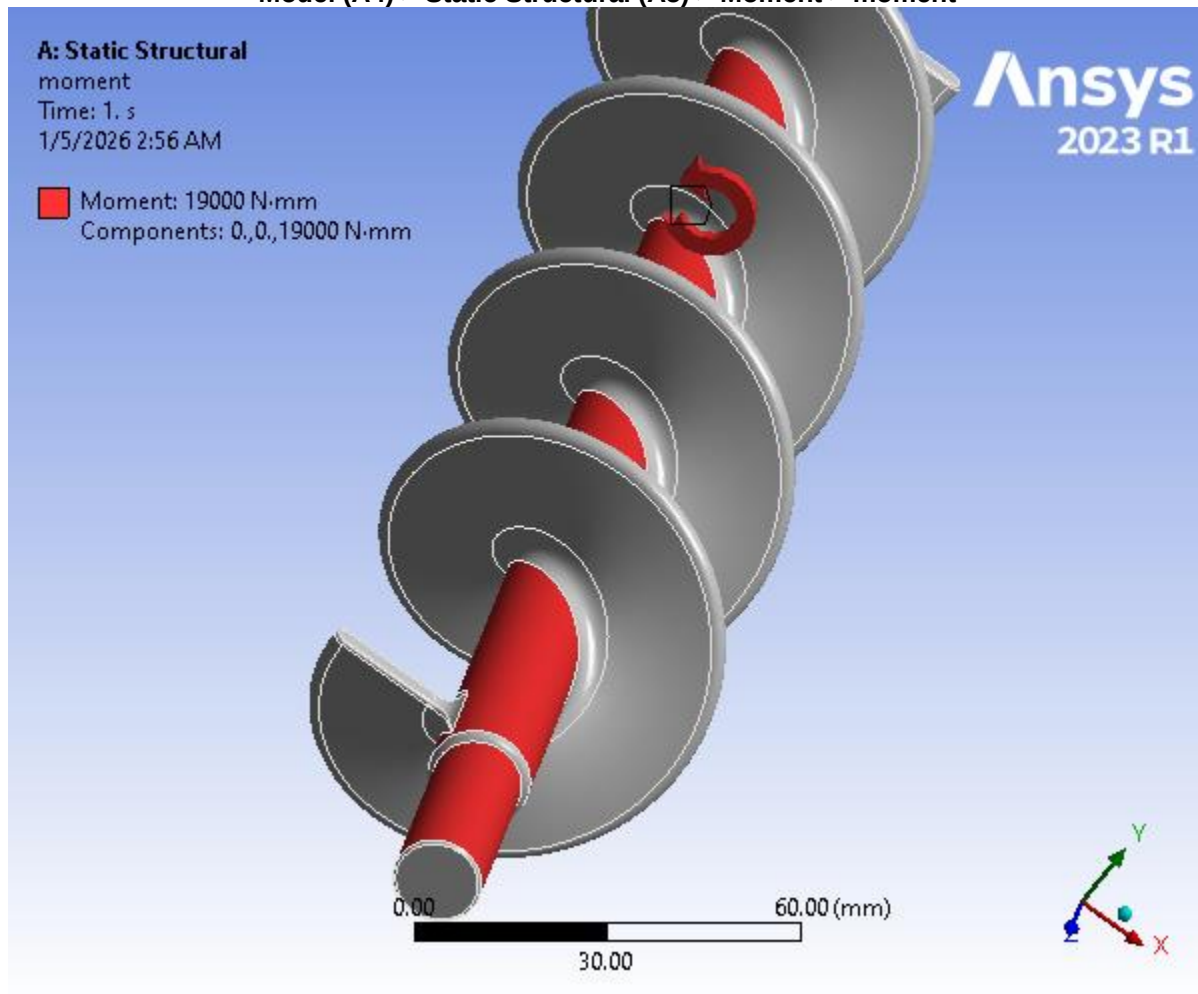


FIGURE 3
Model (A4) > Static Structural (A5) > Moment > moment



Solution (A6)

TABLE 13
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	11 m 0 s
MAPDL Memory Used	20.146 GB
MAPDL Result File Size	1.2305 GB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Total Deformation		Equivalent Stress	
State	Solved			
Scope				
Scoping Method	Geometry Selection			
Geometry	All Bodies			
Definition				
Type	Total Deformation	Equivalent (von-Mises) Stress		
By	Time			
Display Time	First	Last		
Separate Data by Entity	No			
Calculate Time History	Yes			
Identifier				
Suppressed	No			
Results				
Minimum	0. mm	7.6397e-005 MPa		
Maximum	9.5471e-002 mm	136.8 MPa		
Average	2.3477e-002 mm	5.1623 MPa		
Minimum Occurs On	Auger			
Maximum Occurs On	Auger			
Information				
Time	1. s			
Load Step	1			
Substep	1			
Iteration Number	1			
Integration Point Results				
Display Option			Averaged	
Average Across Bodies			No	

FIGURE 4
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

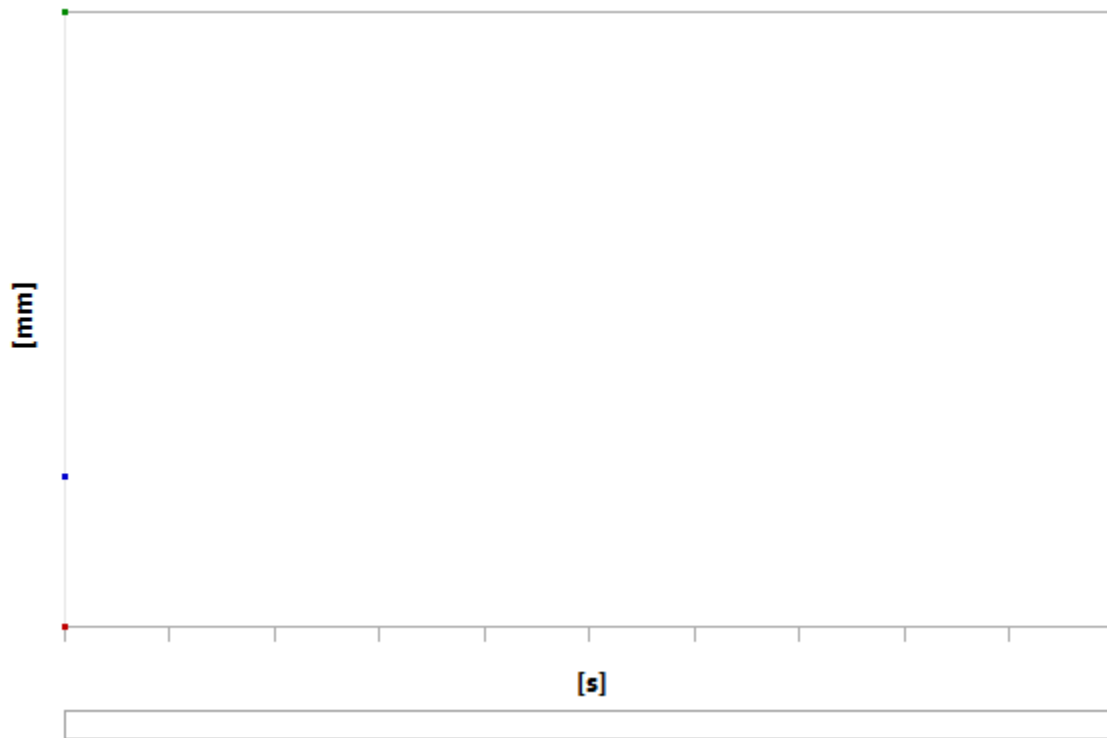


TABLE 16
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1.	0.	9.5471e-002	2.3477e-002

FIGURE 5
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > total deformation

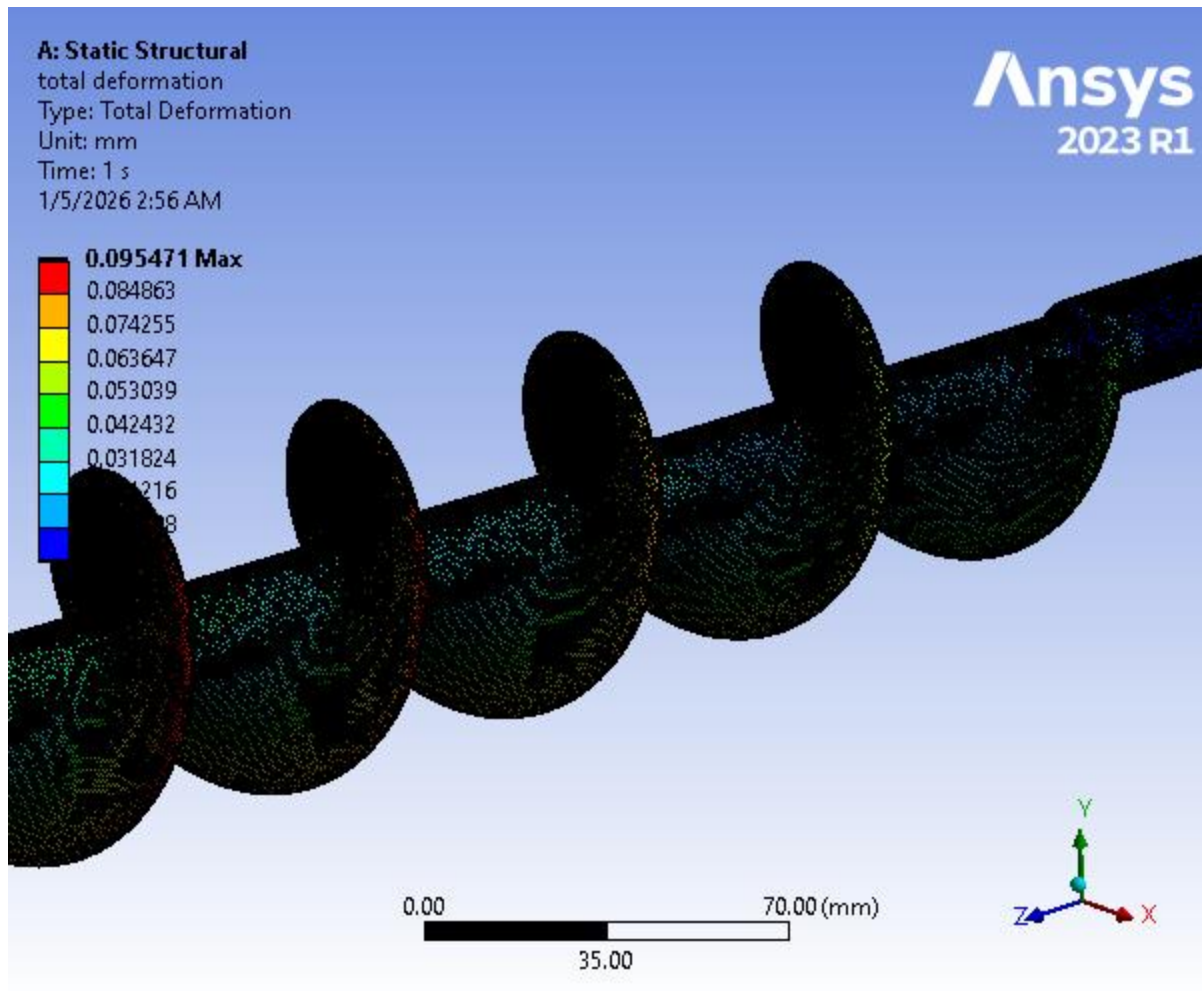


FIGURE 6
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

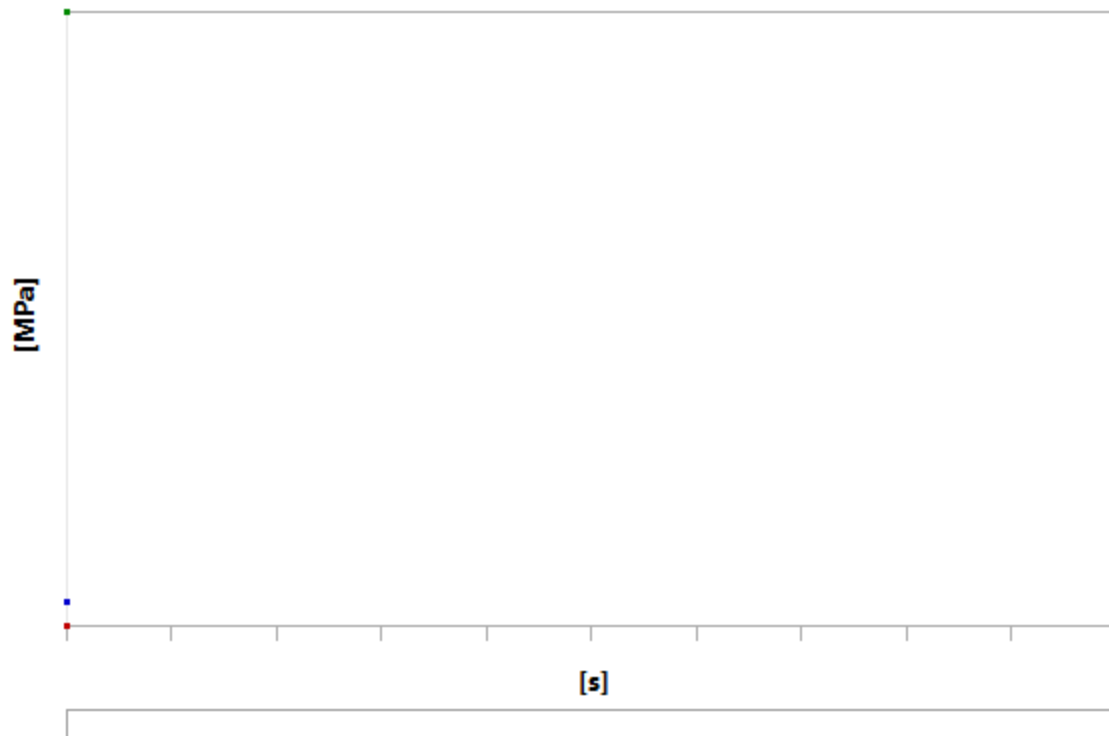


TABLE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1.	7.6397e-005	136.8	5.1623

FIGURE 7
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > equivalent stress

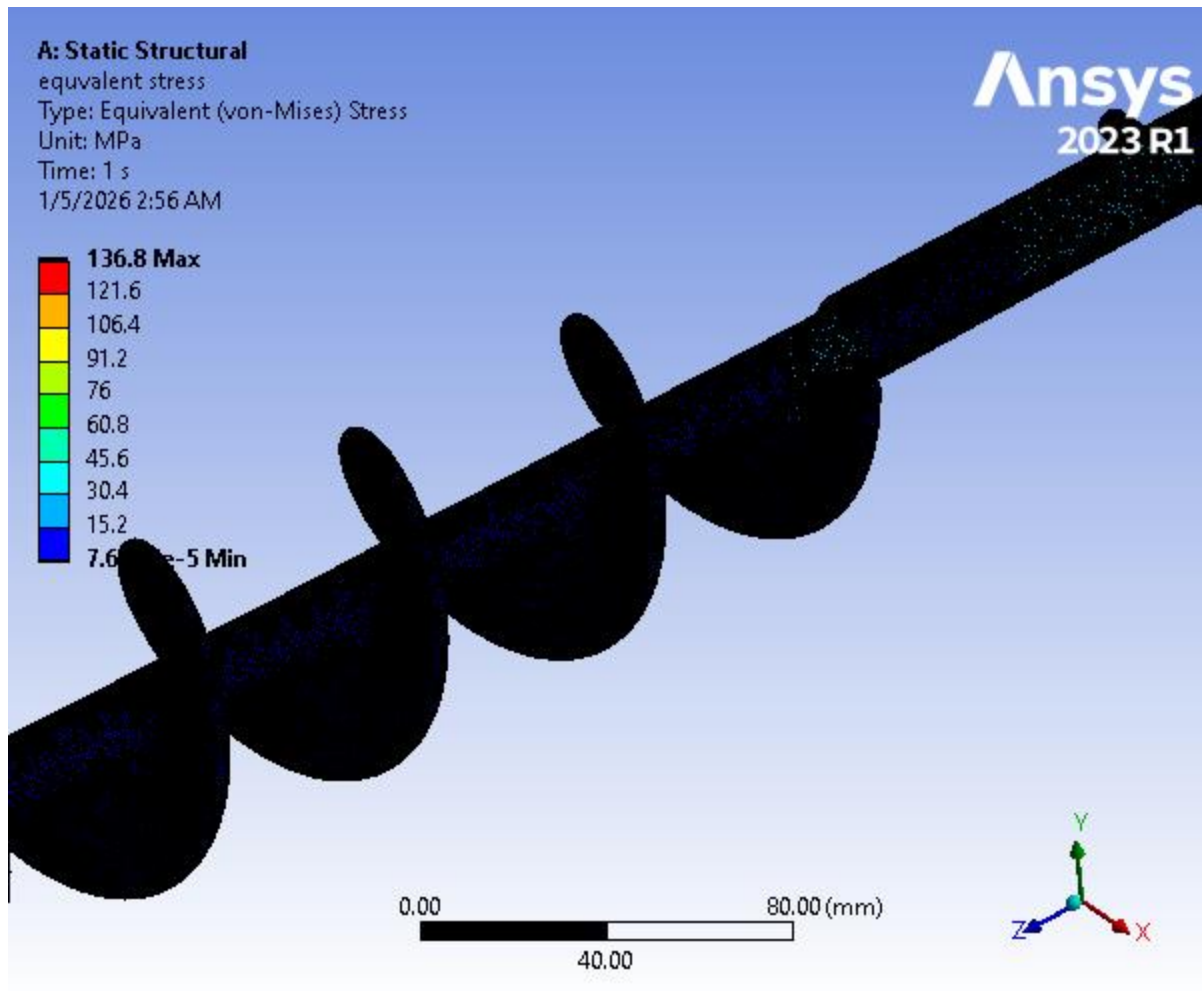


TABLE 18

Model (A4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools

Object Name	<i>Stress Tool</i>
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 19

Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Results

Object Name	<i>Safety Factor</i>
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last

Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	1.5117
Minimum Occurs On	Auger
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 8
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

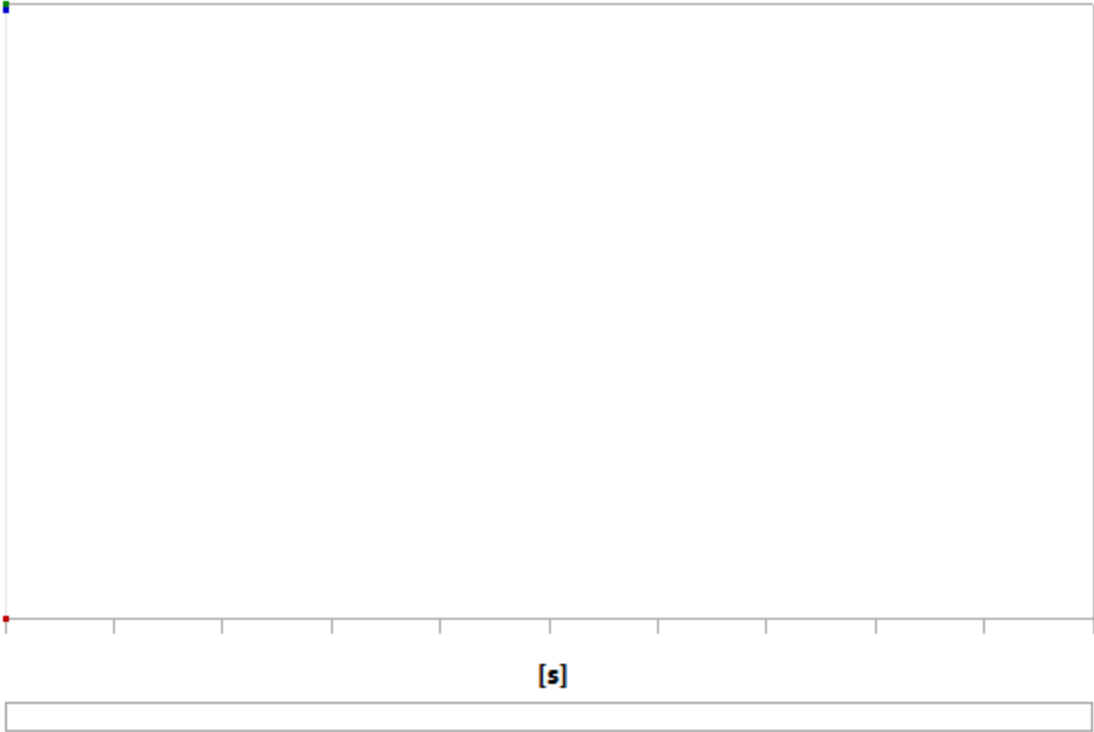
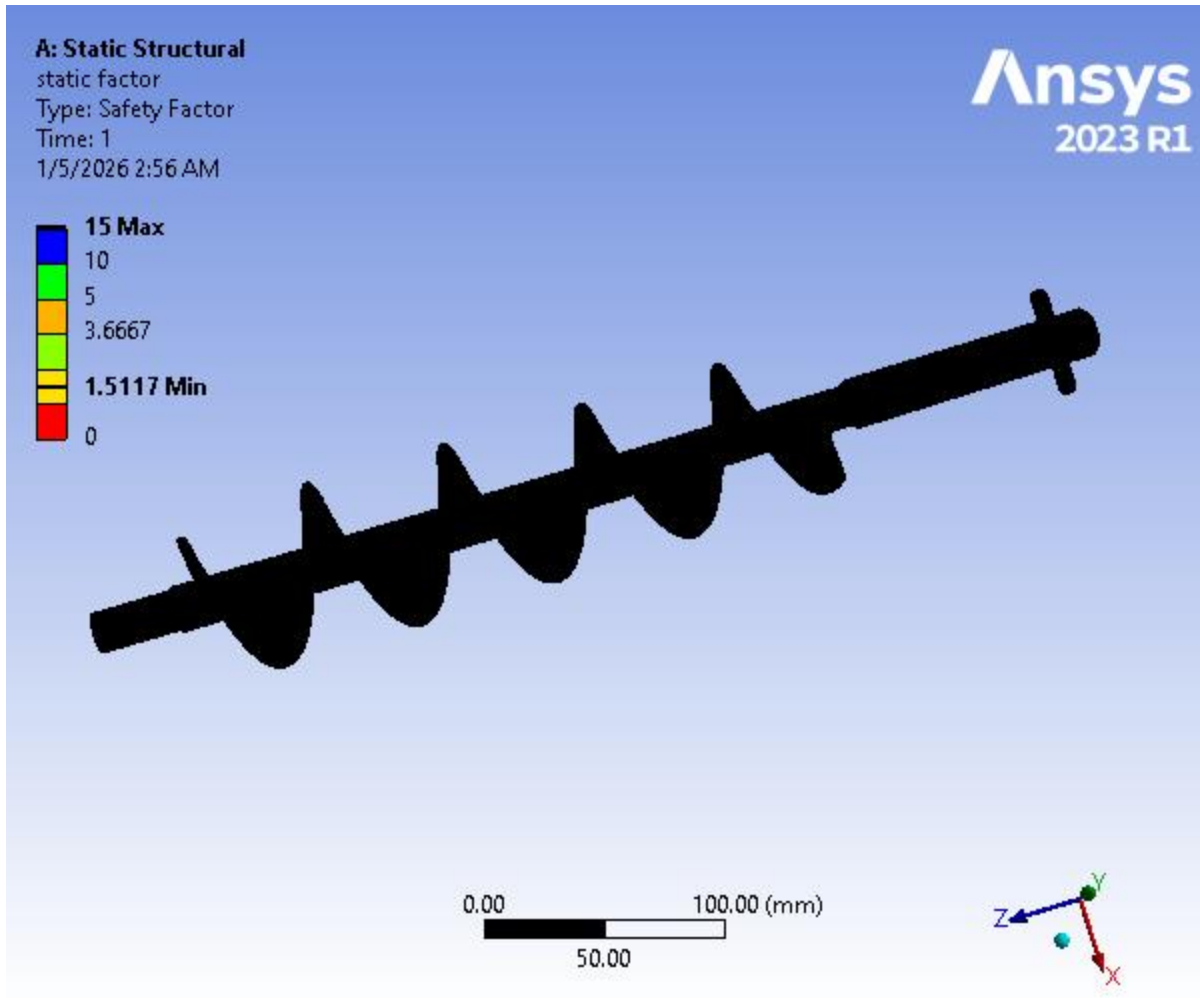


TABLE 20
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	1.5117	15.	14.85

FIGURE 9
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor > static factor



Chart

TABLE 21
Model (A4) > Chart

Steps	End Time [s]
1	1.

Material Data

Stainless Steel

TABLE 22
Stainless Steel > Constants

Density	8.e-006 kg mm ⁻³
Coefficient of Thermal Expansion	1.7e-005 C ⁻¹
Specific Heat	4.8e+005 mJ kg ⁻¹ C ⁻¹
Thermal Conductivity	1.51e-002 W mm ⁻¹ C ⁻¹
Resistivity	7.7e-004 ohm mm

TABLE 23
Stainless Steel > Color

Red	Green	Blue
176	154	176

TABLE 24
Stainless Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 25
Stainless Steel > Compressive Yield Strength

Compressive Yield Strength MPa
207

TABLE 26
Stainless Steel > Tensile Yield Strength

Tensile Yield Strength MPa
206.8

TABLE 27
Stainless Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
517

TABLE 28
Stainless Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 29
Stainless Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
1.935e+005	0.29	1.5357e+005	75000	

TABLE 30
Stainless Steel > Isotropic Relative Permeability

Relative Permeability
1